

ABSTRACT

The research focuses on developing an IoT architecture for battery charging management in response to the increasing demand for electric vehicle (EV) charging infrastructure. As EV adoption rises, integrating renewable energy sources—such as solar power and energy storage systems (ESS) with grid power becomes essential for sustainable charging solutions. The proposed architecture consists of five layers: User Interface Application Layer, Cloud Server and Data Storage, Communication Layer, Microcontroller and Data Acquisition Layer, and Sensor Layer.

This system leverages real-time data monitoring and predictive analytics to optimize power demand, enhance battery management, and improve operational efficiency in EV charging operations. Preliminary simulations indicate a potential operational cost reduction of 90% through optimized resource utilization. The research addresses critical gaps, including the lack of uniform standards for EV charging stations and insufficient studies on the economic viability of vehicle-to-grid technology.

The methodology involves designing a layered architecture that facilitates data remote monitoring, The charging data while integrating multiple energy sources at the Point of Common Coupling (PCC). Future work will focus on refining this architecture to further enhance energy management capabilities, ultimately contributing to a greener transportation ecosystem.

Key words: Solar, ESS – Energy Storage system, Grid, PCC – Point of Coon coupling, Fast Charging, Battery Swapping, Data Visualization, Cloud Integration.